

Control work 1

• Quadrilaterals

The Chapter I assessment: polygons, the four special quadrilaterals, Thales’ theorem and midlines.

LESSON 18

1 × 40 MIN

GEOMETRY 8, ТЕМЫ 13–14

STAGE 9 · UNIT 5 CHECK

LESSON CLOCK

2'

36'

2'

Setting up	2 min
The paper	36 min
Collect in	2 min

LEARNING OBJECTIVES

- Assess the whole of Chapter I in 40 minutes.
- Produce a correctly reasoned proof, not just a numerical answer.
- Draw and label a figure for every question.

TEXTBOOK

National	pp. 33–34
Cambridge	Learner’s Book pp. 103–126
Workbook	Workbook Unit 5

01

Explanation

The paper — 40 minutes, 7 tasks, 14 marks

Two variants, printed below as the Easy and Medium sets. The balance:

TASK	TOPIC	SOURCE	MARKS
1	Angle sum of a polygon	Тема 1	2
2	Angles of a parallelogram	Тема 2	2
3	Rectangle — diagonal or area	Тема 4	2
4	Rhombus or square	Темы 5–6	2
5	Trapezium — height or midline	Темы 7–8, 10–11	2
6	Thales' theorem	Тема 9	2
7	A proof	Темы 2–3	2

Marking note to give before the paper starts: a numerical answer with no figure and no reason earns one mark out of two. Task 7 earns nothing at all without a stated reason at each step — say which property or test you are using.

What to check while they write

- Is there a labelled figure? Learners who skip the figure lose marks on tasks 5 and 7 first.
- On task 6, has the proportion been written before any numbers were substituted?
- On task 7, is a **test** being quoted where a test is needed, rather than a property?

The Hard set below is the work-on-mistakes material: seven pieces of wrong reasoning drawn from the errors this paper produces every year.

02 Worked examples

EXAMPLE 1 Find the error: “ $ABCD$ is a parallelogram with $AC = BD$, therefore it is a rhombus.”

- 1 Equal diagonals is the **rectangle** condition, not the rhombus one.
- 2 The rhombus condition is **perpendicular** diagonals.
- 3 So $ABCD$ is a rectangle.
It is a rhombus only if it is also a square.

ANSWER

A rectangle, not a rhombus.

EXAMPLE 2 Find the error: “ABCD has $AC = BD$, therefore it is a rectangle.”

- ① The word “parallelogram” is missing from the hypothesis.
- ② An isosceles trapezium also has equal diagonals.
A counter-example.
- ③ Equal diagonals alone determine nothing.

ANSWER

Not necessarily a rectangle — it could be an isosceles trapezium.

03 Interactive model

Use this in the work-on-mistakes lesson: show the wrong reasoning, take a vote, then reveal.

Find the flaw in the reasoning

Claimed: in parallelogram $ABCD$, $AC = BD$.

① The parallelogram property is that the diagonals

bisect.

each

other

That

is

AO

=

OC

and

BO

=

OD .

② Nothing forces

AC and

BD to be the same length.

③ Drag any non-rectangular parallelogram and the two diagonals differ.

Equality

needs

a

right

angle.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Figure (diagram)	Chizma	Чертёж
Reason (justification)	Asos	Обоснование
Error	Xato	Ошибка
Prove	Isbotlash	Доказать
Property	Xossa	Свойство
Test (criterion)	Alomat	Признак

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A parallelogram with perpendicular diagonals is:

a rectangle

a rhombus

a trapezium

nothing special

An isosceles trapezium has:

equal diagonals

perpendicular diagonals

diagonals that bisect each other

no special diagonals

The midline of a trapezium with bases 8 and 14 is:

7

11

22

3

To prove a quadrilateral is a parallelogram you need:

a property

a test

either

the angle sum

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. **Variant 1, task 1.** Find the sum of the interior angles of a heptagon.

ANSWER 900°

2. **Task 2.** In parallelogram $ABCD$, $\angle A = 72^\circ$. Find the other three angles.

ANSWER $108^\circ, 72^\circ, 108^\circ$

3. **Task 3.** A rectangle has sides 8 cm and 15 cm. Find the diagonal.

ANSWER 17 cm

4. **Task 4.** A rhombus has diagonals 10 cm and 24 cm. Find its side and area.

ANSWER 13 cm, 120 cm^2

5. **Task 5.** A trapezium has bases 7 cm and 13 cm. Find its midline.

ANSWER 10 cm

6. **Task 6.** $MN \parallel BC$, $AM = 3$, $MB = 6$, $AN = 4$. Find NC .

ANSWER 8

7. **Task 7.** Prove that a quadrilateral whose diagonals bisect each other is a parallelogram.

ANSWER $\triangle AOB \cong \triangle COD$ by SAS $\implies AB = CD$ and $AB \parallel CD \implies$ Test 1.

Medium · the standard question

7 PROBLEMS

1. **Variant 2, task 1.** A regular polygon has exterior angle 30° . Find n and the interior sum.

ANSWER $n = 12, 1800^\circ$

2. **Task 2.** In parallelogram $ABCD$, $\angle A - \angle B = 30^\circ$. Find all four angles.

ANSWER $105^\circ, 75^\circ, 105^\circ, 75^\circ$

3. **Task 3.** A rectangle has diagonal 25 cm and one side 7 cm. Find the area.

ANSWER 168 cm^2

4. **Task 4.** A square has diagonal 10 cm. Find its side and area.

ANSWER $5\sqrt{2} \text{ cm}, 50 \text{ cm}^2$

5. **Task 5.** An isosceles trapezium has bases 5 cm and 17 cm and legs 10 cm. Find the height and area.

ANSWER $h = 8 \text{ cm}, S = 88 \text{ cm}^2$

6. **Task 6.** $MN \parallel BC$, $AM = 4$, $AB = 10$, $AC = 15$. Find AN .

ANSWER 6

7. **Task 7.** Prove that a parallelogram with equal diagonals is a rectangle.

ANSWER Bisecting + equal $\Rightarrow \triangle ABC \cong \triangle BAD$ by SSS $\Rightarrow \angle A = \angle B = 90^\circ$.

Hard · stretch and reason

7 PROBLEMS

1. Find the flaw: “the diagonals of a parallelogram are equal”.

ANSWER They bisect each other; equal needs a rectangle.

2. Find the flaw: “ $AC = BD$, therefore $ABCD$ is a rectangle”.

ANSWER Only if $ABCD$ is already a parallelogram — an isosceles trapezium is a counter-example.

3. Find the flaw: “ $MN \parallel BC$, $AM = 3$, $AB = 12$, so $AN : NC = 3 : 12$ ”.

ANSWER Part-to-whole mixed with part-to-part. Use $\frac{3}{12} = \frac{AN}{AC}$ or $\frac{3}{9} = \frac{AN}{NC}$.

4. Find the flaw: “bases 8 and 14, so the midline is 7”.

ANSWER That is the triangle rule. For a trapezium the midline is the average: 11.

5. Find the flaw: “a rhombus is a rectangle because both are parallelograms”.

ANSWER Both are parallelograms, but neither is a special case of the other — only the square is both.

6. Find the flaw: “opposite sides equal, therefore a rectangle”.

ANSWER Opposite sides equal gives a parallelogram (Test 2); a rectangle needs a right angle or equal diagonals as well.

7. Find the flaw: “three lines evenly spaced by eye cut equal segments, by Thales”.

ANSWER Thales' theorem requires the lines to be **parallel**, which “evenly spaced by eye” does not establish.

07 Homework

After the control work

Re-solve, in full, every task you lost marks on. Quarter II begins with Pythagoras' theorem.

1. Re-solve every task you lost marks on, with a labelled figure and a stated reason.
2. Write out the four errors from the Hard set in your own words.

3. *Revise the properties of a right-angled triangle from Grade 7 — the next chapter opens with Pythagoras.*